
REPORT ON DESIGN

Proposed Information System — Agile Development Approach

Popular Diagnostics Limited, Khulna

Course No: CSE 3120 · Information Systems Design Laboratory

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1. INTRODUCTION

The design phase is widely regarded as the most creative and technically demanding stage of the system development life cycle, as it transforms the findings of the preceding Recognition of Need, Feasibility Study, and Analysis reports into a concrete, implementable blueprint. While the term "design" refers to the final system itself, it equally describes the structured process by which that system is conceived, refined, and prepared for construction.

The earlier analysis of Popular Diagnostics Limited, Khulna, established that the branch's existing Miracle Hospital Information System (MHIS) follows an externally controlled, vendor-driven Iterative Waterfall process — one that is slow to change, closed to local modification, and constrained by proprietary licensing. Rather than proposing a wholesale replacement of this platform, the design presented in this report introduces a complementary, locally developed system, referred to throughout as **DiagnoLink**, which operates alongside the existing MHIS to resolve the structural problems identified during the visit (vendor lock-in, absence of a local server, manual parameter selection risk, limited imaging automation, and the lack of patient-facing digital access).

DiagnoLink is deliberately designed and developed using the Agile Software Development Methodology rather than the rigid Iterative Waterfall approach that governs the vendor's core platform. Agile is selected because it allows the local development team to deliver functional increments — such as the local synchronization engine, the parameter validation layer, and the patient portal — in successive short sprints, each validated directly with branch staff and patients. This stands in deliberate contrast to the vendor's slow, change-request-driven update cycle, and directly addresses the agility and responsiveness gap identified in the Feasibility Study.

2. SYSTEM ARCHITECTURE

The proposed system follows a layered, three-tier architecture consisting of a Presentation Layer, an Application (Service) Layer, and a Data Layer, supplemented by an Integration Adapter that allows DiagnoLink to communicate with the existing MHIS/LIS platform without requiring any modification to its proprietary codebase. This separation ensures that the new system can evolve independently and rapidly, while still respecting the vendor's no-code-sharing restriction.

Users — patients, lab technicians, billing staff, and doctors — interact only with the Presentation Layer, which renders the patient portal and staff dashboards. The Application Layer houses the core business logic, including the parameter validation engine, the imaging integration handler, and the offline synchronization controller. The Data Layer maintains a local, branch-resident database that mirrors and extends the data already present in MHIS, while the Integration Adapter performs one-directional or API-level data exchange with the vendor's central server, ensuring that DiagnoLink never writes directly into the closed MHIS database.

Presentation Layer

Patient Portal · Staff Dashboard · Admin Panel



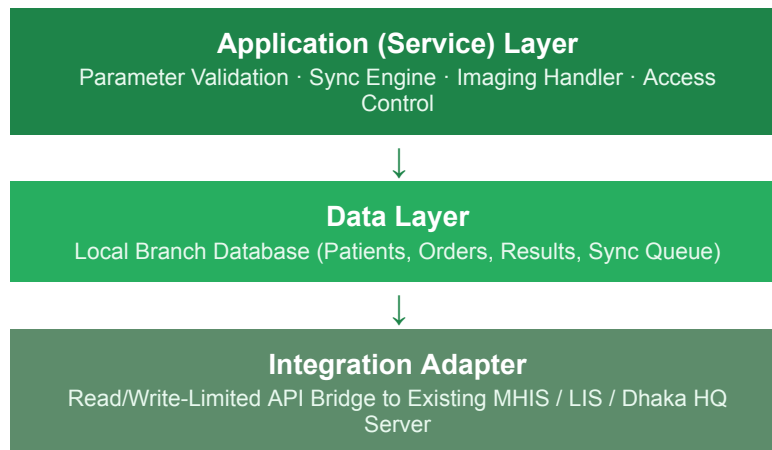


Figure 1: Layered Architecture of the Proposed DiagnoLink System

3. MODULE DESIGN

The system is organized into six major functional modules, each developed and delivered as an independent Agile increment, and each mapped directly to one or more problems identified during the industrial visit.

3.1 Local Resilience & Synchronization Module

This module maintains a continuously updated local copy of patient, billing, and test-order data at the Khulna branch, allowing day-to-day operations to continue uninterrupted even when the connection to Dhaka HQ or the international vendor server is unavailable. Once connectivity is restored, queued transactions are synchronized automatically in the background. This module directly resolves the absence of a branch-level local server, eliminating the single point of failure that previously caused total operational outages during network disruption.

3.2 Parameter Validation Module

Before a test order reaches the LIS machine, this module cross-checks the manually selected parameters against a verified test-order-to-parameter reference table and flags any mismatch for staff review. It does not replace the LIS's own processing but acts as a safeguard layer in front of it, substantially reducing the risk of diagnostic error caused by incorrect manual parameter selection.

3.3 Imaging Integration Module

This module automates the transfer of raw image files from radiology and imaging hardware — including output from the 3D printer used for internal organ visualization — directly into the corresponding patient record. By removing the manual handling step that previously slowed down radiology reporting relative to pathology, this module brings imaging turnaround time closer to parity with the rest of the diagnostic pipeline.

3.4 Patient Portal Module

A secure, patient-facing web and mobile interface allows patients to view their test reports, billing history, and appointment records online. This module directly answers the previously identified gap in digital patient access and supports the personalized health-record initiative that was already in early development at the time of the visit.

3.5 Vendor Liaison & Change Request Module

Recognizing that the core MHIS platform cannot be modified locally, this module provides a structured internal ticketing and tracking system for documenting, batching, and following up on change requests sent to the vendor. While it cannot eliminate the dependency on the Indian vendor team, it significantly improves the visibility and accountability of the change-request process, reducing the practical friction caused by vendor lock-in.

3.6 Data Residency & Compliance Module

This module maintains an encrypted, branch-controlled copy of essential patient data independently of the international vendor server, partially addressing the data sovereignty concerns raised in the Feasibility Study. It also prepares anonymized, standardized data exports in a format compatible with future integration into the Government of Bangladesh's National Health Portal.

DATA FLOW DIAGRAMS (DFD)

The data flow diagrams below decompose the proposed DiagnoLink system across three levels of detail, following the same notation convention used in earlier process-modeling work for this project: external entities are shown as solid rectangles, processes as numbered circles, and data stores as open-ended rectangles.

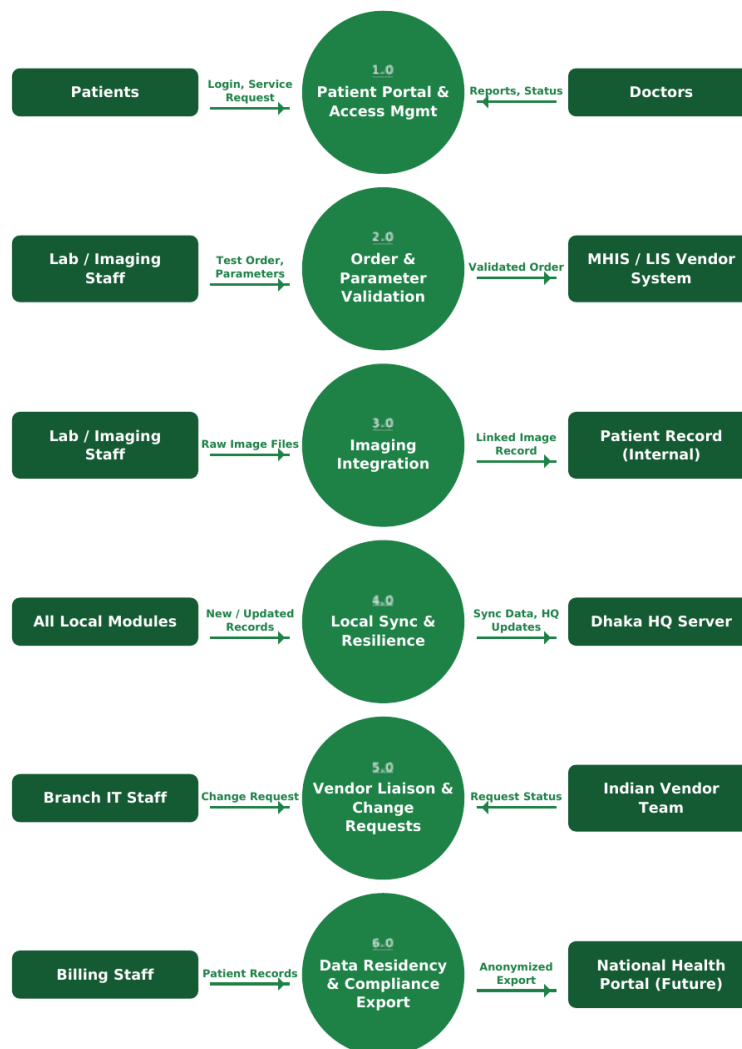
Level-0 DFD

The Level-0 diagram presents the six major processes introduced in Section 3 (Module Design) and maps each one to its relevant external entities, establishing the overall scope of data exchange between users, the local system, and the existing MHIS/vendor infrastructure.

LEVEL-0 DFD — PROPOSED DIAGNOLINK SYSTEM

Popular Diagnostics Limited, Khulna — Main Processes & External Entities

■ External Entity ● Process □ Data Store (Level 1+)



Note: Processes 1.0-6.0 correspond directly to the six modules described in Section 3 (Module Design) of this report. Doctors interact primarily with Process 1.0 and 2.0 to receive reports and review prescriptions.

Figure 3: Level-0 Data Flow Diagram of the Proposed DiagnoLink System

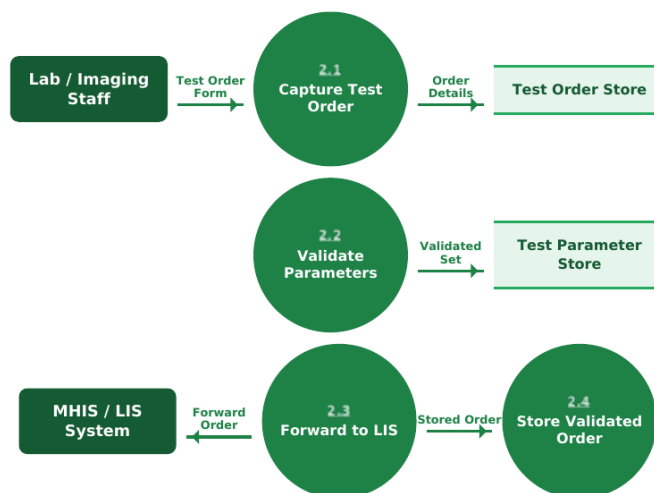
Level-1 DFD

The Level-1 diagram decomposes two of the most critical processes — Order & Parameter Validation (2.0) and Local Sync & Resilience (4.0) — into their constituent sub-processes, introducing the data stores that support them: the Test Order Store, Test Parameter Store, and Sync Queue Store.

LEVEL-1 DFD — PROPOSED DIAGNOLINK SYSTEM

Decomposition of Process 2.0 (Order & Parameter Validation) and Process 4.0 (Local Sync & Resilience)

2.0 ORDER & PARAMETER VALIDATION



4.0 LOCAL SYNC & RESILIENCE

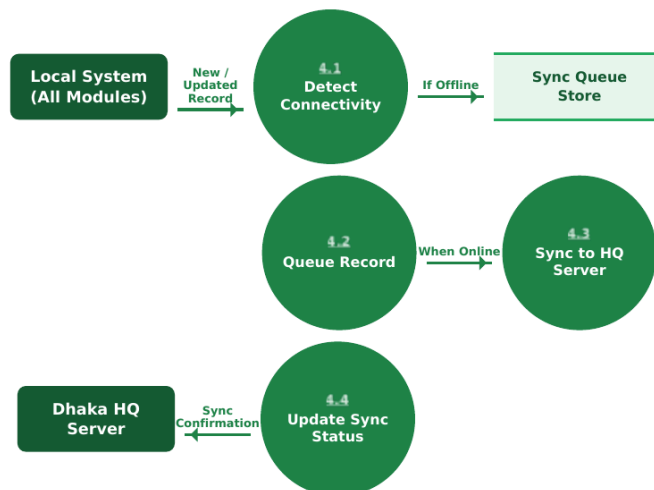


Figure 4: Level-1 Data Flow Diagram — Order & Parameter Validation and Local Sync & Resilience

Level-2 DFD

The Level-2 diagram provides a further breakdown of sub-process 2.2 (Validate Parameters), the component most directly responsible for mitigating the manual parameter selection risk identified during the Analysis phase. It introduces the Parameter Reference Table as a dedicated data store and shows the decision logic that separates an approved order from a flagged mismatch.

LEVEL-2 DFD — PROPOSED DIAGNOLINK SYSTEM

Further Decomposition of Process 2.2 (Validate Parameters) — Addressing the Manual Parameter Selection Risk (P4)

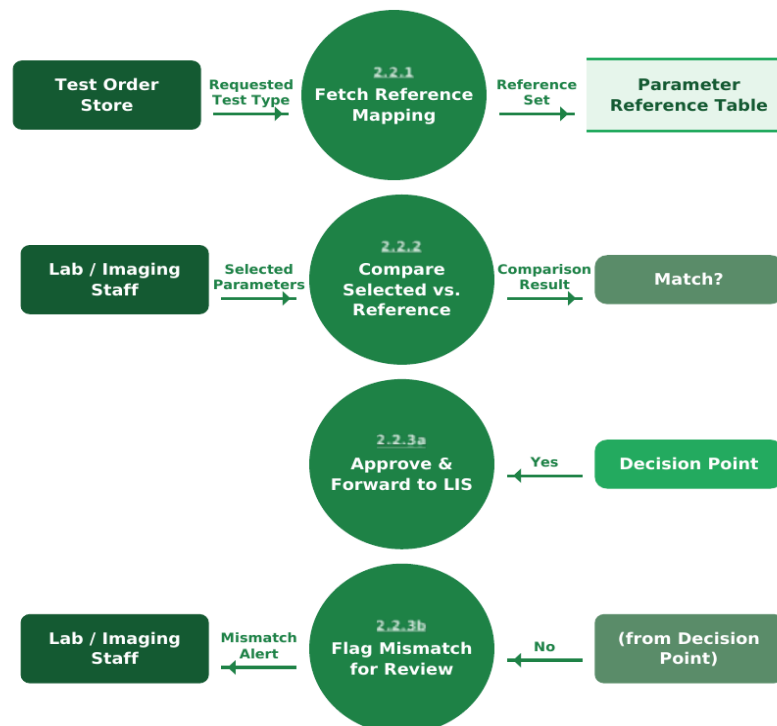


Figure 5: Level-2 Data Flow Diagram — Decomposition of Process 2.2 (Validate Parameters)

This sub-process directly targets the most significant preventable source of diagnostic error identified during the Analysis phase- manual, unverified parameter selection at the LIS machine. By inserting an automatic comparison step (2.2.2) against a maintained parameter Reference Table before any order reaches the LIS, the system flags mismatches for human review rather than allowing incorrect parameters to silently pass through.

4. DATABASE DESIGN

The local database supporting DiagnoLink is structured around seven major entities. It is intentionally kept separate from the MHIS database, communicating with it only through the Integration Adapter, so that the design remains fully compliant with the vendor's closed-codebase policy while still enabling the new functionality described above.

Entity	Purpose	Key Relationship
Patient	Stores patient demographic and contact information.	One-to-many with Test Order; one-to-one with Patient Portal Account
Test Order	Represents each diagnostic request raised at the counter.	Many-to-one with Patient; one-to-many with Test Parameter and Imaging Record
Test Parameter	Stores the validated parameter set applied to a given test order.	Many-to-one with Test Order
Imaging Record	Stores metadata and file references for radiology and imaging outputs.	Many-to-one with Test Order
Sync Queue	Tracks records awaiting synchronization with the Dhaka HQ / vendor server.	References any of the above entities by type and identifier
Patient Portal Account	Manages patient login credentials and access permissions.	One-to-one with Patient
Change Request Log	Tracks vendor change requests raised by branch IT staff.	Independent administrative entity referenced by module documentation

This structure allows the Test Order entity to act as the central hub of the schema, linking patient identity on one side to test parameters and imaging output on the other, while the Sync Queue entity provides a uniform mechanism for tracking the synchronization state of any record type without duplicating logic across modules.

5. USER INTERFACE DESIGN

The user interface design follows a role-based, minimal-interaction philosophy, recognizing that the system's daily users range from highly technical IT staff to non-technical lab assistants, billing clerks, and patients with limited digital literacy. Each interface is built bilingually in Bangla and English to ensure accessibility across all user groups.

Patient Portal

The patient-facing interface presents a simple dashboard summarizing recent reports, outstanding bills, and upcoming appointments. Reports are accessible with a single tap once securely authenticated, removing the previous requirement of a physical branch visit to retrieve results.

Staff Dashboard

The staff interface centers on a single order-intake screen that guides lab technicians through parameter selection with real-time validation feedback, alongside a synchronization status indicator that clearly shows whether the branch is operating in connected or offline mode.

Admin Panel

Administrators access a consolidated panel for managing the change-request log, reviewing compliance export status, and monitoring overall system health, consolidating tasks that were previously scattered across informal communication channels.

Across all three interfaces, the design favors clear visual status indicators, role-restricted views that hide irrelevant functionality, and a consistent layout grammar, which together reduce training time and minimize the likelihood of user error during daily operation.]

6. PROCESS DESIGN

The proposed system overlays a refined process onto the existing patient service flow, intercepting the workflow at the two points identified as most error-prone or fragile during the analysis: parameter selection and post-result data synchronization.

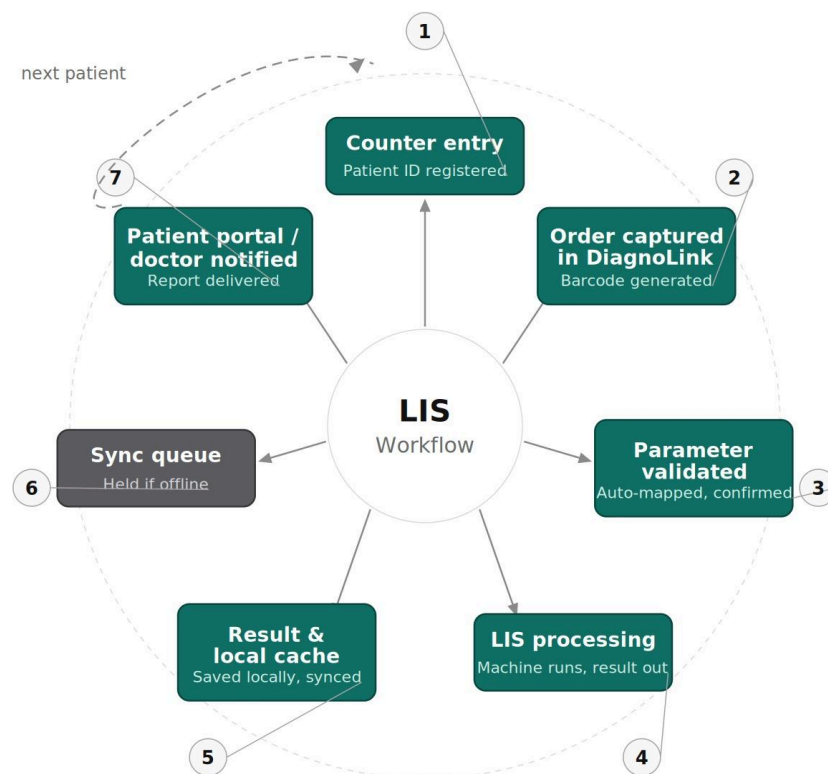


Figure 6: Information Flow Through the Proposed DiagnoLink System

Information entered at the counter is immediately captured by DiagnoLink in parallel with the existing MHIS billing process. Once a sample reaches the laboratory, the Parameter Validation Module reviews the selected test parameters before they are submitted to the LIS machine, reducing the chance of an erroneous result. After the LIS generates the result, it is written to the local cache; if the branch is offline at that moment, the record enters the Sync Queue and is transmitted automatically once connectivity returns. Finally, the patient and attending doctor are notified through the patient portal, closing the loop without requiring a physical visit purely to check status.

7. SECURITY DESIGN

Security is addressed at every layer of the proposed architecture, given the sensitivity of healthcare data and the data sovereignty concerns identified earlier in this project. Authentication is enforced through role-specific login credentials for staff, and a separate one-time-password-based login flow for patients accessing the portal, ensuring that no single shared credential set is used across user categories.

Authorization follows a strict role-based access control model: lab technicians can view and submit test parameters but cannot access billing records; billing staff can view payment information but not raw diagnostic data; and administrators alone can access the change-request log and compliance export functions. Passwords are never stored in plain text; all credentials are stored using industry-standard one-way cryptographic hashing, and portal sessions expire automatically after a period of inactivity.

Data security is reinforced through encryption of the local database both at rest and during synchronization with the Dhaka HQ server, alongside a continuous audit log that records every access to patient records. This combination of access control, credential protection, and encrypted storage directly responds to the data sovereignty and security gaps highlighted during the Feasibility Study.

8. TECHNOLOGY STACK

The technology choices for DiagnoLink are guided by three priorities: compatibility with Agile, sprint-based delivery; suitability for an offline-first, low-connectivity branch environment; and alignment with the technical skill set realistically available to a local development and maintenance team, as recommended in the Feasibility Study.

A modern, MVC-structured web framework such as Laravel is proposed for the application layer, as its built-in routing, authentication scaffolding, and modular structure are well-suited to delivering independent modules — such as the Parameter Validation and Patient Portal modules — across separate Agile sprints without destabilizing the rest of the system. A relational database management system such as MySQL is proposed for the local data layer, chosen for its strong support for structured, relationship-heavy healthcare data and its proven reliability in offline-capable, branch-level deployments.

The presentation layer is built using responsive, mobile-friendly web technologies, ensuring that the patient portal functions equally well on low-end smartphones, which represent the most common access method for patients in this context. Communication between DiagnoLink and the existing MHIS platform is restricted to a lightweight RESTful API bridge, deliberately chosen to avoid any direct dependency on the vendor's internal codebase. Finally, standard version control and Agile sprint-tracking tools support the iterative, incremental delivery model on which the entire design philosophy of this system is based.

9. EXPECTED ADVANTAGES

The proposed design directly resolves each of the structural problems identified during the industrial visit and analysis phases, while also delivering broader improvements across efficiency, accuracy, scalability, maintainability, and usability.

In terms of efficiency, the Local Resilience & Synchronization Module eliminates total service outages caused by network disruption, allowing the branch to continue operating uninterrupted and recovering the productive time previously lost to connectivity failures. Accuracy is improved through the Parameter Validation Module, which directly targets the single largest source of preventable diagnostic error identified in the existing workflow. Scalability is addressed by the modular, Agile-delivered architecture, which allows additional branches or departments to be onboarded incrementally without requiring a full system redesign.

Maintainability is significantly strengthened because DiagnoLink is developed and owned locally, in contrast to the vendor-controlled MHIS platform; future enhancements can be delivered through normal Agile sprints rather than slow, vendor-dependent change requests. Finally, usability is improved through the role-based, bilingual interface design and the introduction of a patient portal, which together reduce training overhead for staff and give patients direct, convenient access to their own health information for the first time.

Taken together, these improvements demonstrate that a locally developed, Agile-delivered complementary system can meaningfully strengthen Popular Diagnostics Limited's information infrastructure without requiring the organization to renegotiate or replace its existing international vendor relationship.